B.Sc. Engg. CSE 1st Semester 13 December 2022

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION **WINTER SEMESTER, 2021-2022**
DURATION: 3 HOURS **FULL MARKS: 150**

CSE 4105: Computing for Engineers

Programmable calculators are not allowed. Do not write anything on the question paper.
Answer **all** 6 (six) questions. Marks of each question and corresponding CO and PO are written in the right margin with brackets.

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1. a) Describe a conceptual file system that achieves the typical storage cataloging purpose – 'Directory contains Directories and Files, File comprises of few storage block(s)'. 7 (CO1) (PO1)

   b) How does a computer boot up? 8 (CO1) (PO1)

   c) A novice user used the commands `ls` and `cd` as shown in Figure 1 and got a message: 'Error: not a directory'.

```
# pwd
/
#ls
User    bin    etc    home    dev    var    myfileDirDev
# cd myfileDirDev
 Error: not a directory
```

Figure 1: Code Snippet for Question 1.c)

   i. What mistake has s/he done? What actions does the user need to perform if s/he wants to investigate the contents of `myfileDirDev`? 3 (CO1) (PO1)

   ii. The user typed `ls -l` in the command prompt. What output s/he will see? (two line sample output will be enough for the demonstration) 2 (CO1) (PO1)

   iii. Now, write a program that will take the output of `ls -l` as input and will print the number of files and directories (you can assume any algorithmic artefact as needed without implementation, for example getLine(), etc.). 5 (CO2) (PO2)

2. a) Describe Hard Disk and its storage mechanism. 10 (CO1) (PO1)

   b) Describe Flash storage technology. How do size and access time vary based on NAND and NOR packaging? 10 (CO1) (PO1)

   c) Answer the following questions: 1×5 (CO1) (PO1)

   i. Why do computers use Cache Memory?
   ii. How many bits make a kilo byte?
   iii. Why don't we use Flash memory as RAM?
   iv. Why is it hard by make more than 1GB processor in a single chip?
   v. What are the components that make a computer system?

3. a) Describe two phishing attack scenarios. (CO1) (PO1)

b) Describe XSS Attack and SQL Injection. Describe how those attacks can be prevented. 8 (CO1) (PO1)

c) Describe Virus and their way of propagations in executables. Describe IAT and ELF (Entry point) Virus. 10 (CO1) (PO1)

a) Which things are patentable? Describe in light of first three sections of "Title 35 of the United States Code". 7 (CO1) (PO1)

b) How do IT organizations generate revenue from patents, sometimes even from the products of other companies? Briefly provide two examples of such revenue generation. 8 (CO1) (PO1)

c) Describe HIPAA and FISMA for Information and Cyber Security. 10 (CO1) (PO1)

a) What are the things/tasks you do when you plan and complete a Tour (visit a new place)? Do you improve yourself in repetitive tours? What things do you expect to improve? Do you find any resemblance of your activities with that of a software development process? 5 (CO1) (PO1)

b) Describe the tasks of compiler, assembler and linker in producing a binary. 10 (CO1) (PO1)

c) Answer the followings: 2×5 (CO1) (PO1)

i. Why are AJAX rich applications more vulnerable to XSS attacks?
ii. Why quick and on demand SQL writing are prone to SQL Injection?
iii. Internet is predominated by world wide web (www). Write two applications or protocols that are not www.
iv. How is a URL resolved by the browser?
v. Can we write a code and generate executable without an IDE? Explain.

a) Describe the task of miner in a block chain. 7 (CO1) (PO1)

b) What is a Merkel Tree in peer-to-peer network? How is the Merkel Tree used in Block chain? 8 (CO1) (PO1)

c) Answer the following questions: 2x5 (CO1) (PO1)

i. How do block chain invalidate double spending?
ii. How is PoW byzantine fault tolerant?
iii. 'Increased hash rate means the block chain is more secured.' – Explain.
iv. What hashing algorithm is used in Bitcoin?
v. Write a simple hashing technique.